

**Akdeniz University · Department of Computer Engineering**  
**CSE 427 — Wireless Sensor Networks · Midterm Examination (Practice)**  
**ANSWER KEY**

**PRACTICE PAPER — not issued by the instructor. Numbers checked with a script; marking scheme is a suggestion.**

**Question 1 — WSN Fundamentals and Application Taxonomy**

*14 points*

- (a) Definitions (4 pts: 1 each definition, 1 each reason)

**Mote:** a small embedded sensor node — sensor(s), MCU, radio, memory, power supply — that senses, processes and communicates wirelessly.

**Sink / gateway:** the collection point that gathers the network's data and bridges it to users or other networks (Internet, database, dashboard).

Not "small Wi-Fi": (1) nodes are battery powered with tiny memory/CPU, so energy dominates every design choice; (2) traffic is small, periodic or event-driven measurement data over lossy multi-hop links, attached to a physical environment — it is measurement infrastructure, not general-purpose connectivity.

- (b) Classification (6 pts: 0.5 per class, 1 per requirement with reason)

| #  | Pattern      | Main need                    | Reason                                                        |
|----|--------------|------------------------------|---------------------------------------------------------------|
| M1 | Periodic     | None critical                | A missing sample is tolerable; the next one comes in a minute |
| M2 | Event-driven | Low latency + reliability    | Plants are damaged if the alarm is late or lost               |
| M3 | Query-driven | None critical                | Operator can simply ask again                                 |
| M4 | Control      | Authentication + reliability | A wrong or forged command changes the physical world          |

- (c) Control and security (4 pts: 2 explanation, 2 attack)

Monitoring only observes: bad data is annoying. Actuation changes physical state: an incorrect or forged command can cause damage, so authentication, integrity and safe defaults become requirements.

**Example attacks on M4: an attacker injects or replays an "open vents" command (spoofing / replay), or jams the channel so the command never arrives.**

**Question 2 — Mote Hardware and Contiki**

*14 points*

- (a) Block diagram (5 pts: 0.5 per block, 1.5 flow + dominant blocks)

Sensors → ADC → MCU → Radio → Antenna, with Memory attached to the MCU and the Power subsystem feeding all blocks.

**Lifetime is dominated by the radio and the power subsystem.**

- (b) Contiki process (5 pts: 1 period, 1 window, 1.5 static, 1.5 busy-wait)

**"sample" is printed every 5 seconds, visible in Cooja's Mote Output window.**

**static:** protothreads do not keep a stack across `PROCESS_WAIT_EVENT_UNTIL`; a local (automatic) variable would lose its value while the process is suspended.

**No busy wait:** the process yields; the MCU can enter a low-power mode until the timer event arrives. A busy loop keeps the CPU active (wasting energy) and blocks the other processes in the cooperative scheduler.

- (c) Sky vs Cooja mote (4 pts)

**Sky mote:** emulates the real MSP430 hardware and CC2420 radio and runs the compiled firmware — closer to hardware timing, memory limits and energy; slower.

**Cooja mote:** the Contiki code compiled natively and run at software level — much faster, but not hardware-faithful (no real memory/timing constraints).

Choose Sky for labs that measure timing/energy or must match hardware (the course default); Cooja motes for quick logic tests or large networks.

**Question 3 — Radio and Physical Layer**

*18 points*

- (a) Received power at 20 m (4 pts)

$$PL(20) = 40 + 30 \cdot \log_{10}(20) = 40 + 30 \cdot 1.301 = 79.03 \text{ dB}$$

$$P_r = 0 - 79.03 \approx -79.0 \text{ dBm}$$

(b) Maximum range (5 pts)

$$40 + 30 \cdot \log_{10}(d) \leq 94 \Rightarrow \log_{10}(d) \leq 1.8$$

$$d_{\max} = 10^{1.8} \approx 63.1 \text{ m}$$

(c) Shadowing (5 pts: 2 probability, 3 distance)

At  $d_{\max}$  the mean margin is 0 dB, so the link works iff  $X_\sigma \leq 0$ .

$$P(\text{link works at } 63.1 \text{ m}) = 0.5$$

$$\text{fade margin} = z_{0.90} \cdot \sigma = 1.28 \cdot 4 = 5.13 \text{ dB}$$

$$30 \cdot \log_{10}(d) \leq 94 - 40 - 5.13 = 48.87 \Rightarrow d = 10^{1.629}$$

$$d_{0.90} \approx 42.6 \text{ m}$$

(d) RSSI / LQI / PRR (4 pts)

**RSSI**: received signal strength (dBm) measured by the radio. **LQI**: the radio's link-quality indicator, based on the chip error rate of the received frame (802.15.4). **PRR**: fraction of sent packets that are received.

The **transitional region** is the distance band between reliable connectivity and no connectivity, where PRR varies widely and unpredictably because of fading, shadowing and noise. Links are **asymmetric** because nodes differ in transmit power, sensitivity, noise floor and local interference, so A may hear B while B barely hears A.

#### Question 4 — Energy

20 points

(a) Always listening (4 pts)

$$T = 2500 \text{ mAh} / 21.5 \text{ mA} = 116.3 \text{ h}$$

$$\approx 116 \text{ h} \approx 4.8 \text{ days}$$

(b) Duty-cycled (8 pts: 2 per term, 2 lifetime)

$$f_{RX} = 0.01, f_{TX} = 0.004 / 60 = 6.67 \cdot 10^{-5}, f_{\text{sleep}} = 1 - f_{RX} - f_{TX} \approx 0.98993$$

| State       | Fraction             | Current   | Contribution     |
|-------------|----------------------|-----------|------------------|
| RX / listen | 0.01                 | 21.5 mA   | 0.2150 mA        |
| TX          | $6.67 \cdot 10^{-5}$ | 19.2 mA   | 0.0013 mA        |
| Sleep       | 0.98993              | 0.0051 mA | 0.0050 mA        |
| Average     |                      |           | <b>0.2213 mA</b> |

$$T = 2500 / 0.2213 = 11\,295 \text{ h}$$

$$\approx 471 \text{ days} \approx 1.29 \text{ years} \text{ — about } 97\times \text{ longer than (a)}$$

(c) Dominant term (4 pts)

**Listening dominates: 0.215 of 0.221 mA  $\approx$  97 % of the average current.**

Transmissions are negligible; the cost is idle listening. Optimization should reduce listening time — shorter/rarer wake-ups (RDC/MAC design, e.g. ContikiMAC), fewer rendezvous — rather than shrinking packets.

(d) Energest (4 pts)

Energest counts how long (in timer ticks) each component spends in each state: CPU active, LPM, radio transmit, radio listen.

$$E = \sum_{\text{state}} (\text{ticks}_{\text{state}} / \text{ticks-per-second}) \cdot I_{\text{state}} \cdot V$$

Currents come from the datasheet (e.g. CC2420, MSP430);  $V \approx 3 \text{ V}$  for two AA cells.

#### Question 5 — Medium Access Control

20 points

(a) Hidden / exposed terminal (6 pts: 3 each)

**Hidden**: A and C both send to B. A and C cannot hear each other, so carrier sensing says "free" to both; their frames collide at B.

**Exposed:** B sends to A while C wants to send to D. C hears B, senses the channel busy and defers — although its transmission to D would not disturb A. Capacity is wasted.

(b) B-MAC vs X-MAC (6 pts: 2 B-MAC, 2 X-MAC, 2 overhearing)

**B-MAC:** the preamble must be at least as long as the wake-up interval so the receiver surely sees it: the sender transmits for the full **500 ms** every time.

**X-MAC:** the sender repeats short strobes carrying the destination address; the receiver answers with an early ACK as soon as it wakes. With a uniformly random wake-up time the expected wait is  $T_w/2$ .

**B-MAC: 500 ms · X-MAC: 250 ms on average (500 ms worst case)**

Overhearing: a non-target node that wakes up decodes one short strobe, sees it is not the destination and goes back to sleep, instead of listening to a whole long preamble.

(c) TDMA graph coloring (5 pts: 3 assignment, 2 proof)

A, B, C form a triangle, so at least 3 slots are needed. A 3-slot assignment:

| Node | A | B | C | D | E | F |
|------|---|---|---|---|---|---|
| Slot | 1 | 2 | 3 | 1 | 1 | 2 |

Check: D is adjacent to B(2), C(3), F(2); E to C(3), F(2); F to D(1), E(1) — no conflict.

**Minimum = 3 slots**

(d) TSCH channel (3 pts: 2 channel, 1 reason)

$$(1003 + 2) \bmod 4 = 1005 \bmod 4 = 1 \Rightarrow \text{seq}[1]$$

**Channel 20**

Hopping gives frequency diversity: narrow-band interference and multipath fading on one channel only hit a fraction of transmissions.

**Question 6 — 6LoWPAN and RPL Routing**

14 points

(a) Why adaptation (4 pts)

An 802.15.4 frame carries at most 127 bytes (and much less after MAC headers), while IPv6 has a 40-byte header and requires links to carry 1280-byte packets.

**6LoWPAN uses header compression and fragmentation / reassembly.**

(b) Fragmentation (5 pts: 2 payload sizes, 3 count)

first:  $102 - 4 = 98 \rightarrow$  round down to multiple of 8  $\rightarrow$  96 bytes

later:  $102 - 5 = 97 \rightarrow$  96 bytes; remaining  $400 - 96 = 304 = 3 \cdot 96 + 16$

**5 fragments carrying 96, 96, 96, 96, 16 bytes**

(c) RPL parent choice (5 pts: 2 choice, 1 hop count, 2 ETX reason)

via A:  $1.2 + 2.0 = 3.2$  · via B:  $2.5 + 1.0 = 3.5$

**X chooses A; path ETX = 3.2**

Hop count need not agree: B's path ETX of 1.0 suggests B is one hop from R, so hop count would prefer B (2 hops) over A.

ETX counts expected transmissions including retries, so it reflects real link quality and energy cost; hop count prefers long, lossy links that need many retransmissions.